MAPLE & MAIN · MEDIA INVESTMENT

What the mix model can tell you, what it cannot, and what to test

A full marketing mix model on three years of weekly data, scored against the truth, with a budget optimiser that is judged on whether it would actually have worked.

Synthetic data. No client, employer or proprietary information.

EXECUTIVE SUMMARY

The model fits beautifully and cannot be trusted on its own. Six tests make it worth acting on.

$65M to $159M

spread in media's measured contribution across equally good versions of the same model. The data alone cannot choose between them.

109%

average error in channel return on investment with no in-market test. After tests that measure the shape of each curve, 7%.

$523k

a year from reallocating the same budget, once the curves are calibrated. Acting on the uncalibrated model instead would have *lost* $258k.

Recommendation: treat the mix model as the allocation engine and in-market tests as its instrument. Run holdouts at two spend depths on every channel over a year, then move the budget: less paid search, less display, more flyer, more email.

APPROACH

Built the standard way, then stress-tested in ways a client rarely sees

156 weeks of sales against $25.2M of media across six channels, 2023-11-06 to 2026-10-26.

1. Shape the media

Carry-over and saturation searched per channel on out-of-time error, not assumed.

63 candidates per channel

2. Fit the structure

Sales as a base that media multiplies, so contributions add to sales exactly.

Positive effects only

3. Break it

Refit from several honest starting points and see whether the answer holds.

Identification test

4. Calibrate

In-market holdouts pin the channels the calendar hides.

Simulated geo tests

5. Decide

Optimise the budget, then score the recommendation on the true curves.

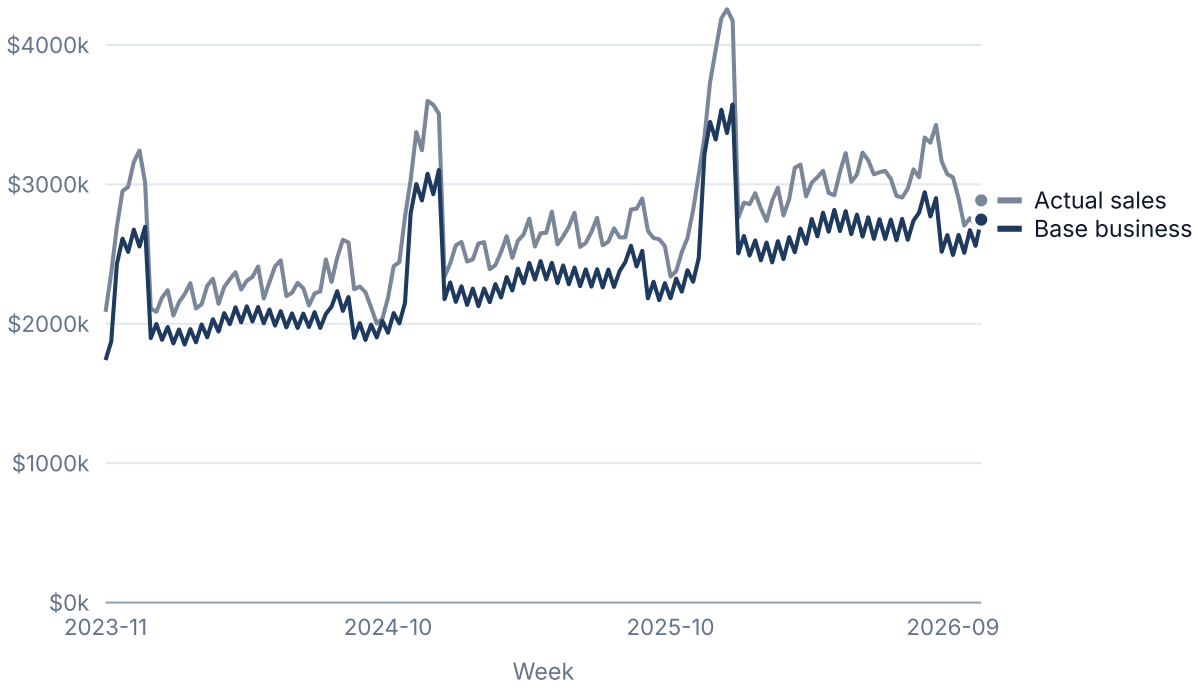
The honest test

DATA USED	CONTROLS IN THE BASE	HOW RESULTS WERE VALIDATED
Weekly sales from every order line, media spend by channel, the trading calendar and realised price against list	Trend, three annual harmonics, holiday peak, back-to-school, discount depth	Four expanding-window folds, multi-start identification, and the response curves the data was generated from

THE MODEL

Sales explained week by week, with media separated from the calendar

Weekly sales: base business and media contribution



The gap between the two lines is what media produced. The base is what the model attributes to trend, season, holidays and price.

Fit	Out-of-time error	Residual pattern
0.937 R^2	**3.5%**	**1.56** DW

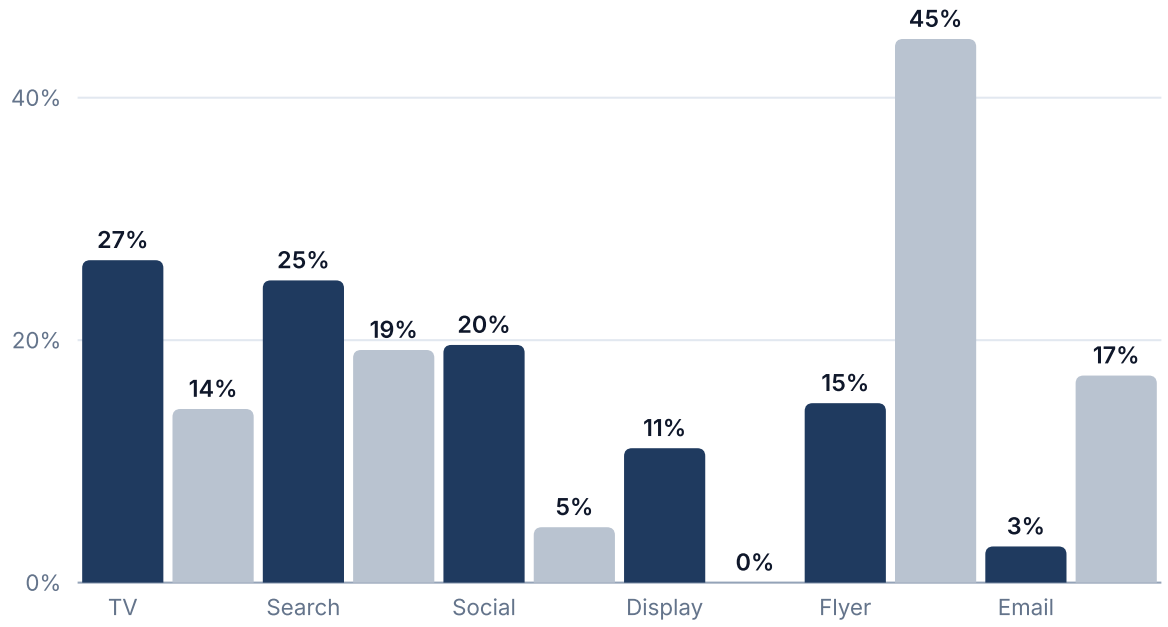
CHANNEL	CARRY-OVER	HALF-SATURATION
TV	70%	$24k
Paid Search	0%	$20k
Paid Social	70%	$18k
Display	80%	$9k
Flyer	0%	$16k
Email	0%	$2k

Carry-over is the share of a week's pressure that survives into the next. Half-saturation is the weekly spend at which a channel delivers half its maximum effect.

THE STANDARD OUTPUT

Contribution and return by channel: the slide every mix model ends on

Share of spend against share of media-driven sales



Dark: share of the media budget. Light: share of the sales media produced, as the uncalibrated model sees it.

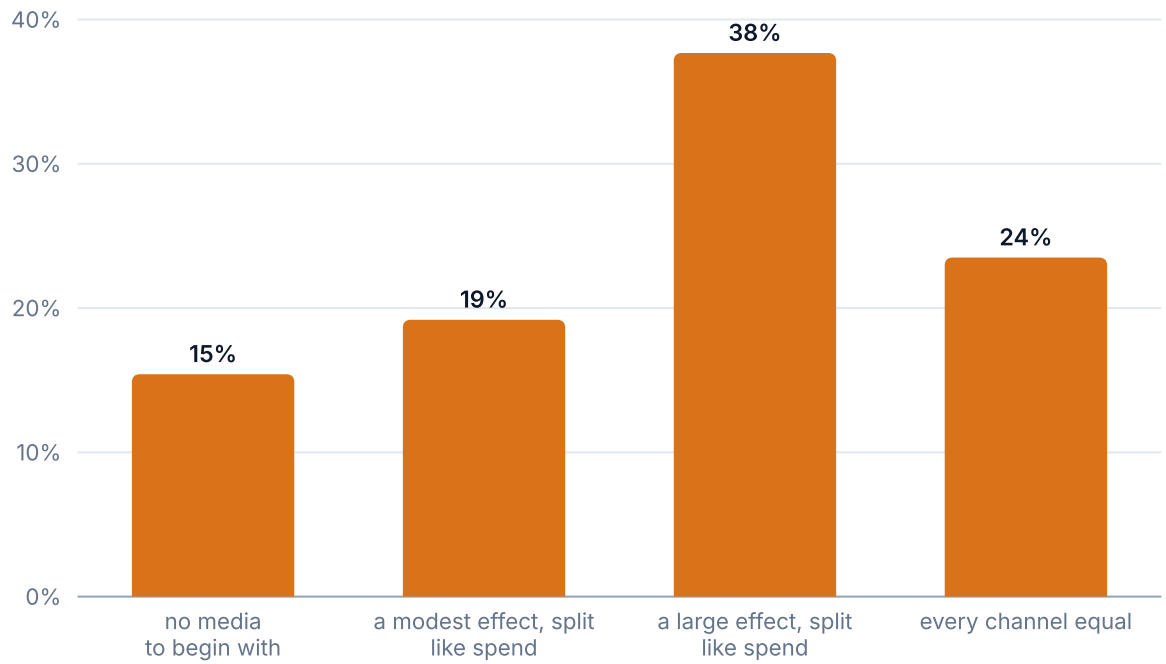
CHANNEL	SPEND	RETURN	NEXT DOLLAR
TV	$6713k	$1.72	$-0.12
Paid Search	$6293k	$2.46	$0.01
Paid Social	$4949k	$0.74	$0.02
Display	$2799k	$0.00	$0.00
Flyer	$3736k	$9.68	$5.98
Email	$753k	$18.30	$1.67

Read this way, the answer is obvious: cut TV and Display, pour everything into Flyer and Email. **That recommendation would have destroyed value.** The next four slides explain why.

THE PROBLEM

Four versions of the same model, fitting equally well, disagree by 2.4 times

Media's share of sales, by where the fit was started



Identical data, identical transforms, identical controls. Only the starting point differs.

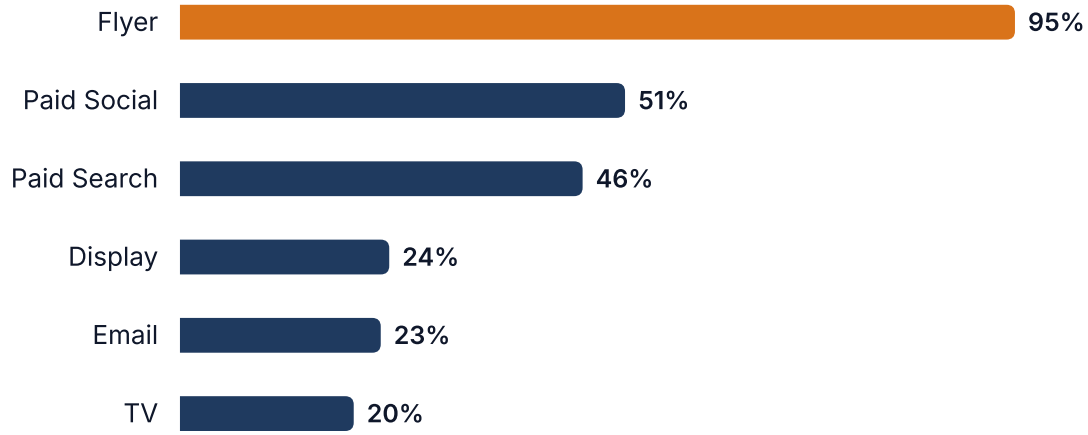
STARTING ASSUMPTION	MEDIA CONTRIBUTION	FIT
no media to begin with	$65M	0.940
a modest effect, split like spend	$81M	0.939
a large effect, split like spend	$159M	0.930
every channel equal	$99M	0.938

The fit statistic cannot tell these apart: 0.93 points of R-squared separates the best from the worst. A mix model reports one of these answers and shows a confidence interval around it that reflects none of this.

WHY

Media is planned around the calendar, so the calendar can explain the media

How much of each channel's pattern the calendar already explains



R-squared of each channel's saturated pressure on the trend, season, holiday and price terms.

Flyer drops are scheduled around the same peaks the model already controls for. **95%** of the flyer variable is reproducible from the calendar alone, so the model cannot tell whether December sales came from the flyer or from December.

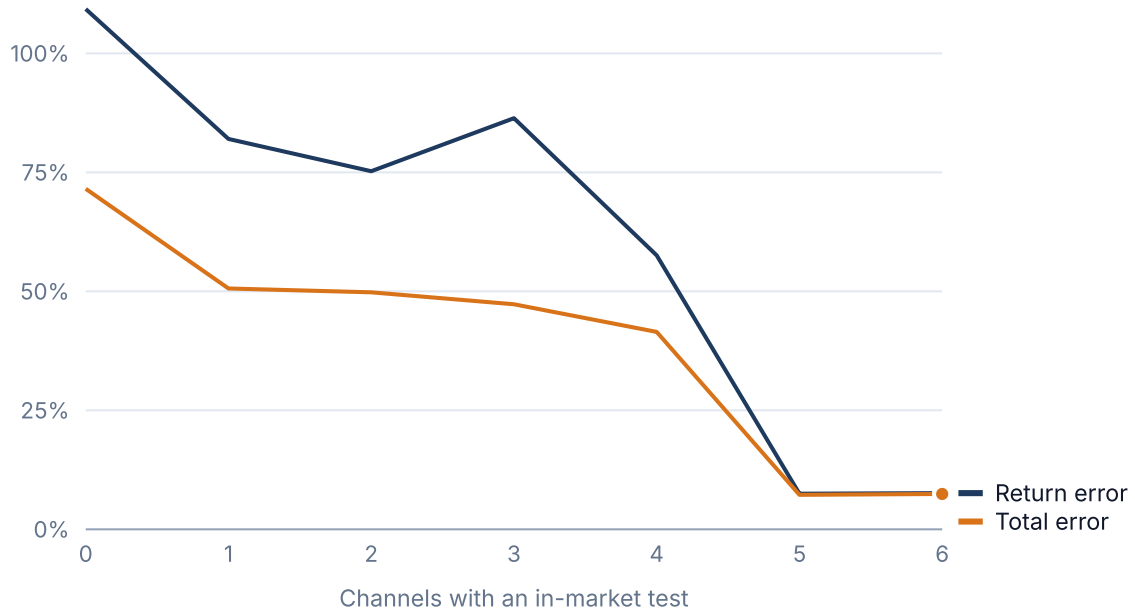
TV is the opposite: it runs in flights that cut across the season, so only 20% of it is calendar. It is the one channel the data can speak to.

> This is the test that should decide where to spend research money: **test the channels the calendar can imitate**, not the ones with the biggest budgets.

THE FIX, PART ONE

Each in-market test buys a measurable amount of accuracy

Average error in channel return, by number of channels tested



Channels are tested in order of how badly the calendar hides them. Error is measured against the true response curves.

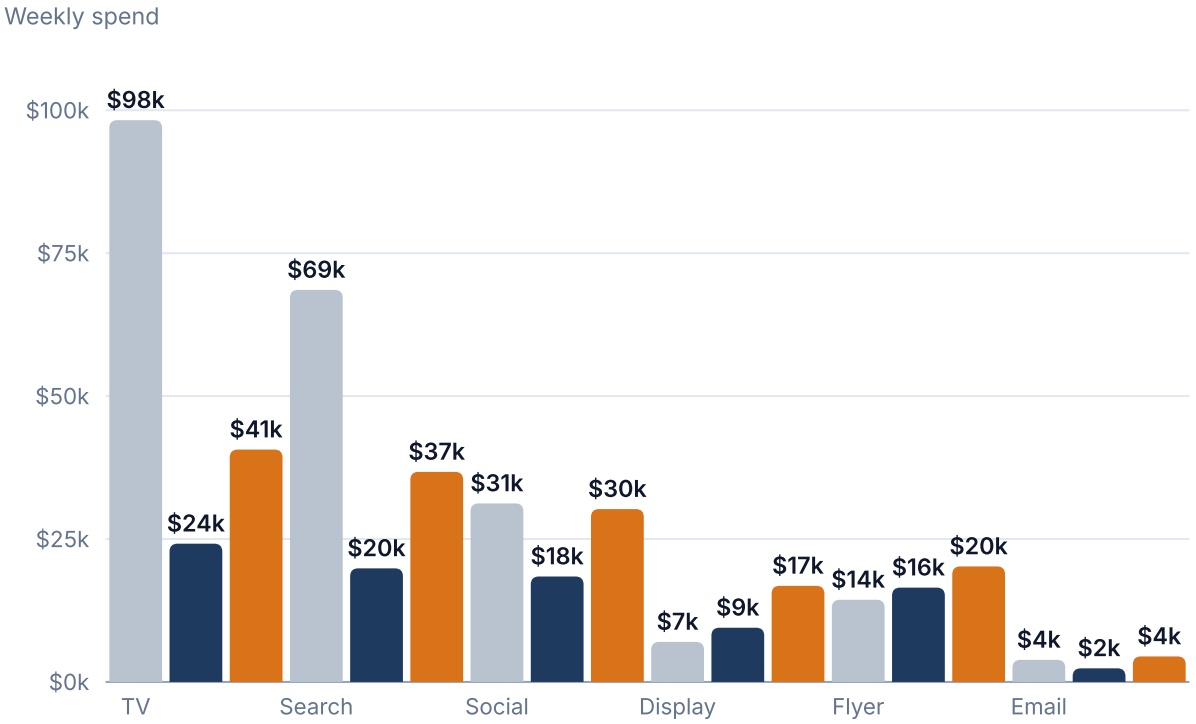
TESTS	CHANNEL ADDED	RETURN ERROR	TOTAL ERROR
0	none	109%	+72%
1	Flyer	82%	+51%
2	Paid Social	75%	+50%
3	Paid Search	86%	+47%
4	Display	58%	+41%
5	**Email**	**7%**	**+7%**

Two tests cut the error in channel returns by a third. Five bring the model within 7% of the truth. The fit statistic barely moves throughout: accuracy and fit are different things.

THE FIX, PART TWO

Knowing what a channel is worth today says nothing about what happens if you move it

Where each channel saturates: model, calibrated model, and truth



Weekly spend at which a channel delivers half its maximum effect. The marker is today's spend.

A single holdout measures one point on a curve. Reallocation depends on the *slope* at that point, which one measurement cannot give.

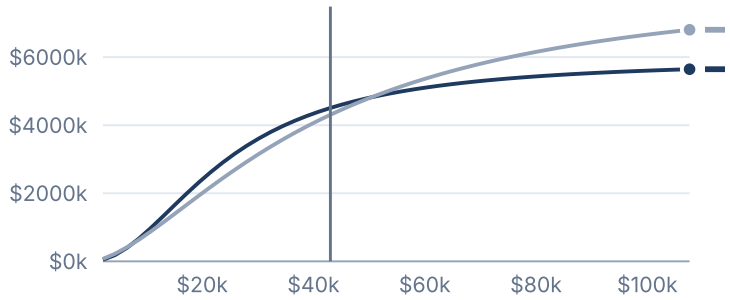
The model alone puts TV's saturation at $98k a week against a true $41k, so it believes TV has far more headroom than it does.

> Holdouts run at **two different depths** on the same channel measure two points, which fixes both the height and the shape of the curve. It is the same test budget spent differently.

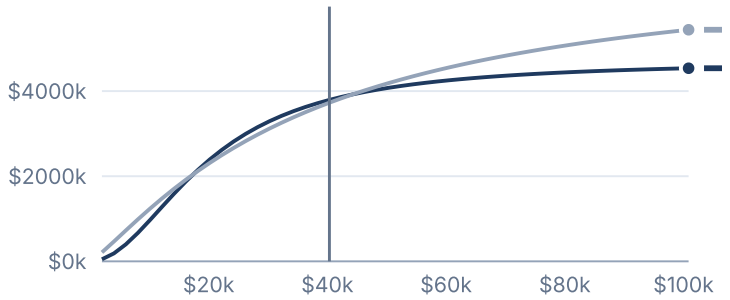
RESPONSE CURVES

The curves that drive every budget decision

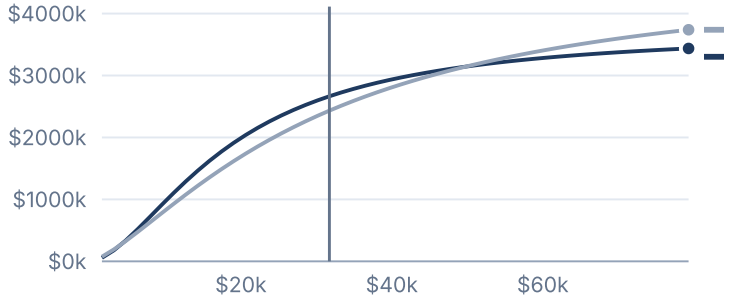
TV



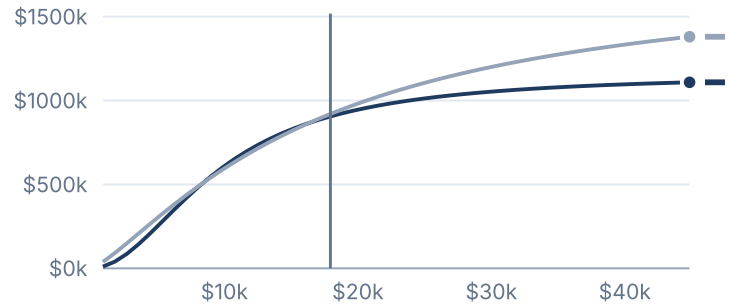
Paid Search



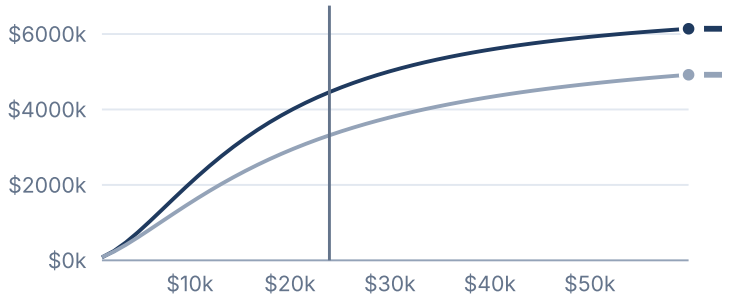
Paid Social



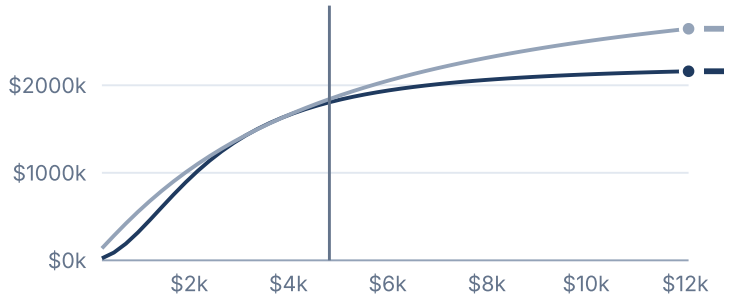
Display



Flyer



Email

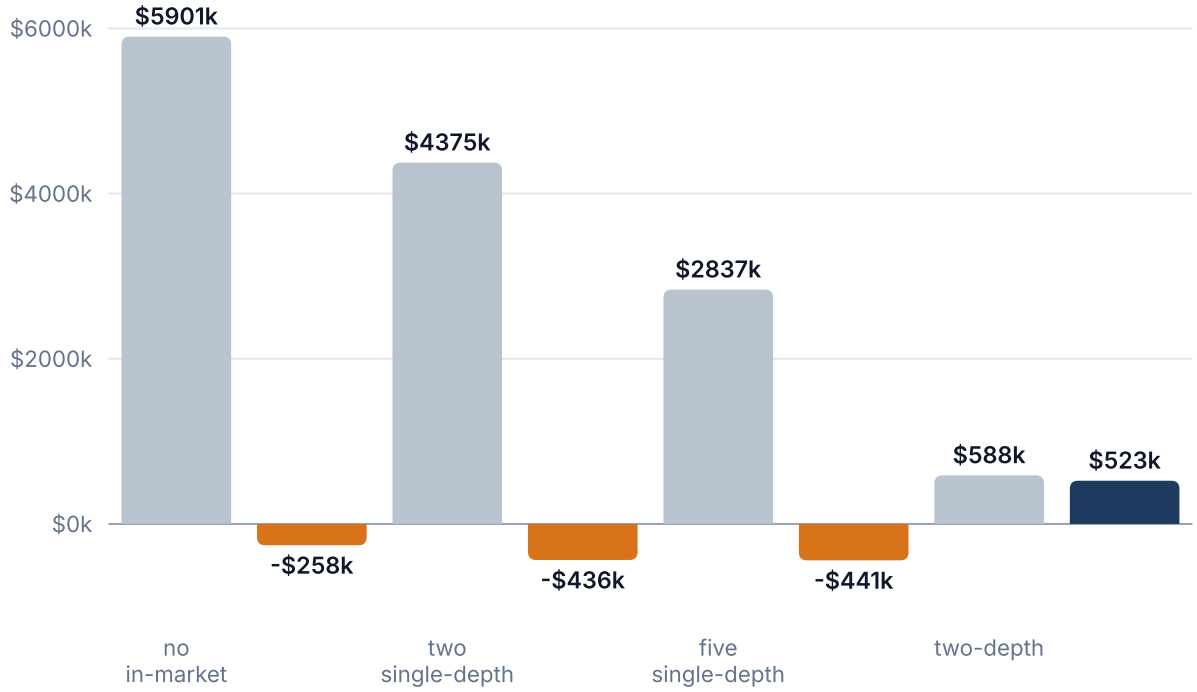


Annual incremental sales at a constant weekly spend. Navy: model plus two-depth tests. Grey: the truth. Amber marker: today's spend. The model alone is left out for readability; it sits far above both.

THE DECISION

Every plan promises a gain. Only one delivers it.

What each version of the model promised, and what it would really have earned



Annual incremental sales from moving the same budget, judged on the true response curves. No budget is added in any plan.

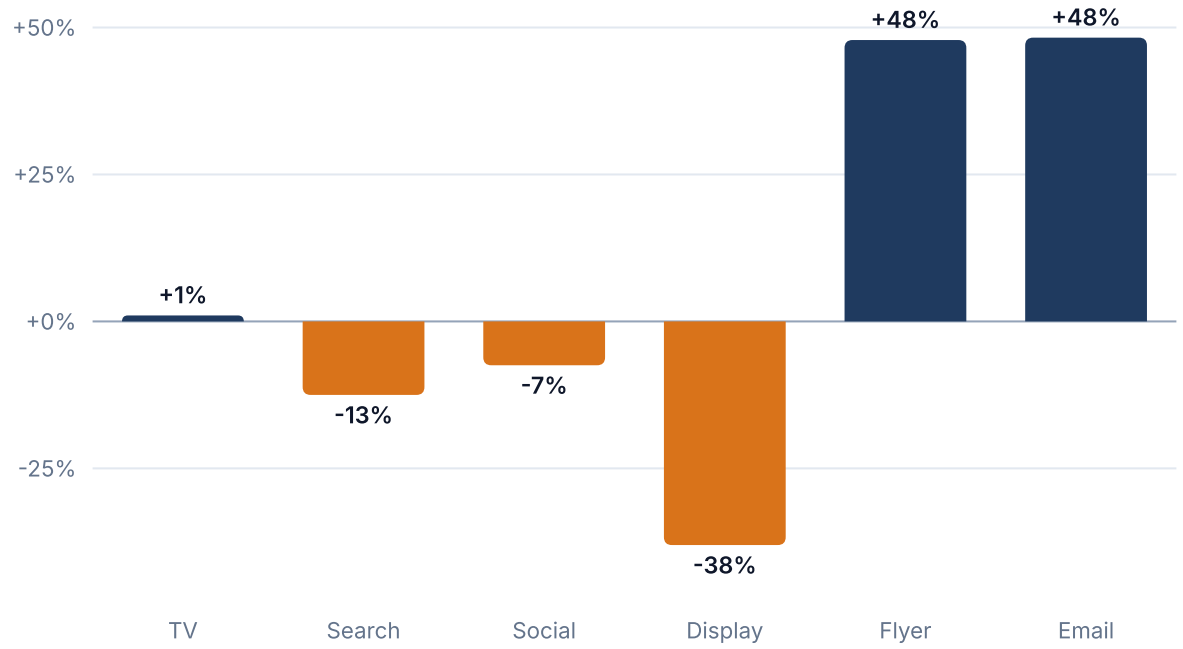
PLAN	PROMISED	ACTUAL
Model alone, no in-market test	$5901k	-$258k
Model plus two single-depth tests	$4375k	-$436k
Model plus five single-depth tests	$2837k	-$441k
Model plus two-depth tests on every channel	**$588k**	**$523k**
Best possible, knowing the true curves	n/a	$630k

The uncalibrated model promised $5.9M and would have lost $258k. The version calibrated on two-depth tests promised $588k and delivers $523k, which is 83% of everything that was available.

THE RECOMMENDED PLAN

Where the money should move

Change in weekly spend by channel



Within a limit of half a channel's current spend in either direction, on an unchanged total budget of $8.4M a year.

CHANNEL	NOW, A WEEK	PROPOSED	RETURN
TV	$43k	$43k	$1.90
Paid Search	$40k	$35k	$1.76
Paid Social	$32k	$29k	$1.60
Display	$18k	$11k	$0.96
Flyer	$24k	$35k	$2.75
Email	$5k	$7k	$7.01

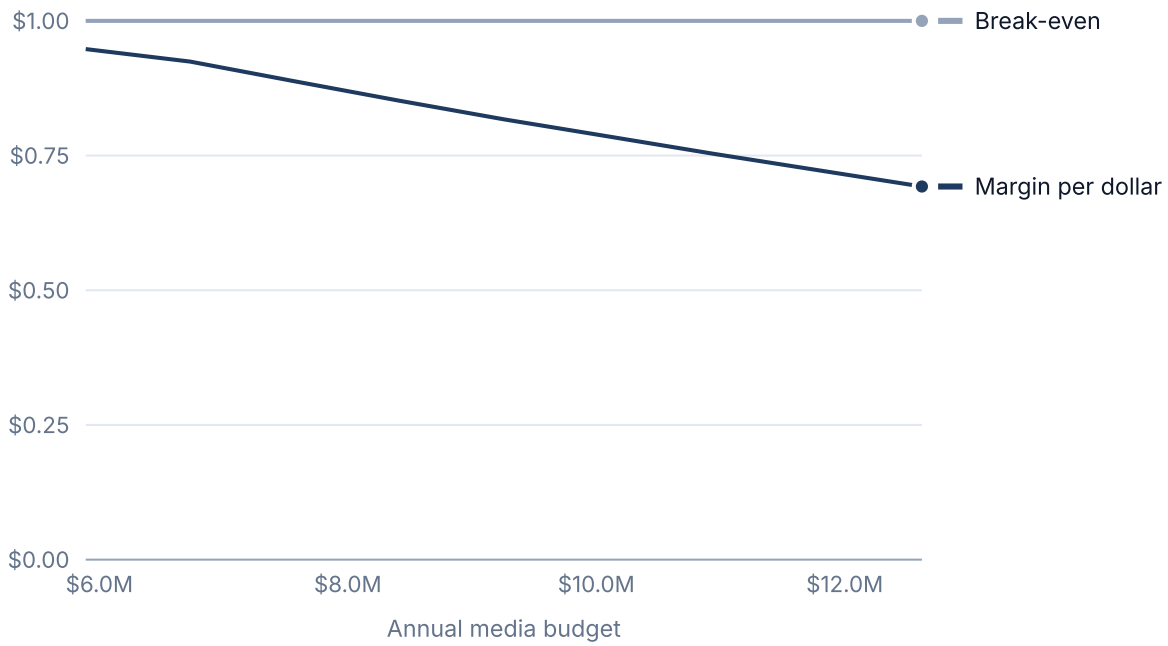
Return is incremental sales per dollar from the calibrated model; the true value is shown in the appendix.

> Every channel is already past the half-saturation point, between 59% and 67% of its maximum effect. There is no channel with large untapped headroom, which is why the gain is a reallocation of 3.2%, not a doubling.

BUDGET SCALE

On this quarter's margin alone, the media budget roughly breaks even

Return on the media budget, by budget level



Incremental gross margin per dollar of media, at the best allocation for each budget level, on the true curves. One is break-even.

At today's budget, media returns **$0.83** of gross margin per dollar spent in the same period. Cutting the budget raises that ratio and lowers total profit; adding to it does the reverse.

That is the wrong frame on its own. Media also recruits customers, and the acquisition work in this portfolio values a new customer at **$428** of first-year margin, with more beyond it.

> Judge the **mix** on the immediate return, which is what this model measures well once calibrated. Judge the **scale** on lifetime value, which needs the acquisition and retention models, not this one.

RECOMMENDATIONS

Four actions

1 **Move the budget as set out on slide 12**

An unchanged total, reallocated on calibrated curves. Review after one quarter against a holdout region so the move itself is measured.

$523k a year

2 **Run holdouts at two spend depths, not one**

Same test budget, twice the information. Start with the channels the calendar can imitate: Flyer and Paid Social.

Turns a 109% error into 7%

3 **Stop reporting a single number for media contribution**

Publish the range the data actually supports alongside the point estimate, and mark which channels are evidence-based and which are inferred.

Prevents the 2.4 times error

4 **Separate the mix question from the scale question**

This model answers where the next dollar goes. Whether there should be more dollars depends on customer lifetime value, which the acquisition and retention models carry.

Correct use of the tool

APPENDIX

Method, assumptions and limits

Model. Weekly sales = base x (1 + sum of channel lifts). The base is log-linear in trend, three annual harmonics, holiday and back-to-school flags, and realised price against list. Channel lift is a Hill saturation of geometrically adstocked spend. Channel effects are constrained to be positive.

Transforms. Carry-over (nine values), saturation shape (five) and half-saturation point (seven) searched per channel by coordinate descent on four expanding-window folds.

Prior. Without evidence to the contrary the split of contribution is pulled gently toward the split of spend, which stops the fit collapsing onto one channel.

Tests. Holdouts are simulated from the true response curves with the sampling error a geo test of thirteen weeks would carry. In a real engagement these are read off the test, not generated.

Accuracy against truth. Final model: average error in channel return 7%, total media contribution +6%.

CHANNEL	MODEL RETURN	TRUE RETURN
TV	$1.90	$1.78
Paid Search	$1.76	$1.74
Paid Social	$1.60	$1.45
Display	$0.96	$0.97
Flyer	$2.75	$2.37
Email	$7.01	$7.20

Limits. Weekly national data cannot see competitor activity, distribution changes or creative quality, none of which exist in this dataset either. Price is controlled but not modelled as a response. The optimiser assumes steady weekly spend and ignores flighting, minimum buys and contractual commitments.

Not tested. Interaction between channels, long-term brand effects beyond the carry-over window, and the effect of media on the customer base rather than on the week's sales.

Synthetic data. Maple & Main is fictional. Its media response curves are known, which is the only reason the accuracy claims above can be made at all. Figures illustrate the method, not any real company.

Code. SQL, Python and SAS, with a model card and QC log, are in the project repository.